No of Pages : 3 **Course Code : 23MX11**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, MARCH 2025

MCA Semester: 1

23MX11 MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE

Time: 3 Hours **Maximum Marks: 100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. **Statistical tables are permitted.**

1. a) Show that a one-to-one function *f:* {1, 2, 3} → {1, 2, 3} must be onto. (3)

b) i) Use Principle of Inclusion and Exclusion to find, how many numbers below 100 are divisible by 2, or 3, or 5? (5)

ii) Prove that $((A - B) - (B - C))^c = A^c \cup B$ using truth table. (5)

c) i) Prove that 7 divides $2^{4n+2} + 3^{2n+1}$ for all nonnegative integers n using mathematical induction. (6)

ii) Let L be the set of all lines in a plane and R be the relation in L defined as R = {(L1, L2): L1 is perpendicular to L2}. Show that R is symmetric but neither reflexive nor transitive. (6)

2. a) Obtain the dual of the following LPP.

Maximize $z = 5x_1 + 6x_2$

subject to $x_1 + 2x_2 = 5$, $-x_1 + 5x_2 \geq 3$, $4x_1 + 7x_2 \leq 8$,

x_1 unrestricted in sign, $x_2 \geq 0$. (3)

b) i) Obtain an optimum basic feasible solution to the following transportation problem using UV method. (7)

Factory	Warehouse				Factory Capacity
	W_1	W_2	W_3	W_4	
F_1	19	30	50	10	7
F_2	70	30	40	60	9
F_3	40	8	70	20	18
Warehouse requirements	5	8	7	14	

ii) Joe Klyne's four children, John, Karen, Mary, and Terri, want to earn some money to take care of personal expenses during a school trip to the local zoo. Mr. Klyne has chosen four chores for his children: mowing the lawn, painting the garage door, cleaning the kitchen and washing the family cars. To avoid anticipated sibling competition, he asks them to submit (secret) bids for what they feel is fair pay for each of the four chores. The understanding is that all four children will abide by their father's decision as to who gets which chore. The following table summarizes the bids received. Based on this information, how should Mr. Klyne assign the chores? One child should be assigned to only one chore.

		Chore 1	2	3	4
Child	1	\$1	\$4	\$6	\$3
	2	\$9	\$7	\$10	\$9
	3	\$4	\$5	\$11	\$7
	4	\$8	\$7	\$8	\$5

(5)

c) Use **simplex** method to solve the following LPP

Maximize $z = 4x_1 + 10x_2$

Subject to $2x_1 + x_2 \le 50$

$2x_1 + 5x_2 \le 100$

$2x_1 + 3x_2 \le 90$

$x_1, x_2 \ge 0$ (10)

3. a) Find whether the given below statements are true or false. Defend your answer with justification

i. Weighted mean is always lesser than arithmetic mean

ii. P(B/A) = P(B) if A and B are mutually exclusive

iii. The daily rainfalls (in mm) of a city for a week are 110, 108, 109, 110, 107, 108, 109, 111. This data reveals that the rainfall is inconsistent (3)

b) i) Assume that the word 'offer' occurs in 80% of the spam messages in my account. Also, let's assume 'offer' occurs in 10% of my desired e-mails (non-spam). 30% of the received e-mails are considered as a scam and remaining are non-spam. Suppose if I will receive a new message which contains 'offer', what is the probability that it is spam? (5)

ii) For the joint probability distribution of two random variables X and Y

Y / X	1	2	3	4
1	4/36	3/36	2/36	a
2	1/36	3/36	3/36	2/36
3	5/36	1/36	1/36	1/36
4	1/36	2a	1/36	5/36

Compute the value of **a** and find the marginal distributions of X and Y (5)

c) i) The frequency distribution for the length, in seconds, of 100 telephone calls was:

Time (seconds)	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100	100 – 120	120 – 140	140 – 160	160 – 180
Frequency	0	5	7	14	28	21	13	9	3

Compute mean, median and standard deviation. (12)

(OR)

ii) Let X be a continuous random variable with probability density function (12)

$$f(x) = \begin{cases} ax & 0 \le x \le 1 \\ a & 1 \le x \le 2 \\ -ax+3a & 2 \le x \le 3 \\ 0 & \text{elsewhere} \end{cases}$$

1) Determine the constant a
2) Compute $P(X \le 1.5)$
3) Compute E(X)

4. a) The average growth of a certain variety of pine trees is 10.1 inches in three years. A biologist claims that a new variety will have improved three-year growth. A random sample of 25 of the new variety has an average three-year growth of 10.8 inches and a standard deviation of 2.1 inches. Test this against the biologist's claim with 0.1 level of significance. (3)

b) The following table shows the 12 month air ticket sales of an Airlines company. Perform exponential smoothing of the data with $\alpha = 0.2$ and forecast the sales for 13th month. (7)

Time t	1	2	3	4	5	6	7	8	9	10	11	12
Sales (in 1000s)	43	41	54	45	46	43	49	42	52	46	51	55

c) Given the following data which relates to a study made by a retail merchant to determine the relation between his sales and weekly advertising expenditure. The following data was recorded by him for a period of 12 days.

Cost x	40	20	25	20	30	50	40	20	50	40	25	50
Sales y	385	400	395	365	475	440	490	420	560	525	480	510

i) Plot a scattered diagram

ii) Find the equation of the regression line to predict weekly sales from the advertising cost

iii) Estimate weekly sales if the advertising cost is $35

iv) Find the correlation coefficient, assess the statistical significance of the value and interpret your results (15)

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